

Project Title: Student Success Predictor – Machine Learning Project Brief

Project Overview

This project challenges you to apply your understanding of Python and core machine learning concepts to predict whether a student will pass or fail based on a variety of academic and personal features. You will perform data exploration, preprocessing, modeling, and evaluation using real-world datasets.

Objective

Build and evaluate supervised learning models to predict student academic success using publicly available datasets.

Datasets (Available Online)

You will use one of the following datasets:

1. UCI Student Performance Dataset

- Link: <https://archive.ics.uci.edu/ml/datasets/student+performance>
- Description: Academic performance data for secondary school students including demographics, school-related attributes, and exam scores.

2. OpenUniversity Learning Analytics Dataset (OULAD)

- Link: https://analyse.kmi.open.ac.uk/open_dataset
- Description: Anonymized data on student learning behavior in an online course environment.

Tasks

1. Data Exploration

- Load and inspect the datasets.
- Visualize distributions, correlations, and patterns.

2. Preprocessing

- Handle missing values.
- Encode categorical variables.
- Normalize numerical features if needed.

3. Feature Engineering

- Create new informative features (e.g., average previous grades).
- Apply feature selection techniques and optional dimensionality reduction.

4. Modeling

- Split data into training and testing sets.
- Train multiple classification models (e.g., Logistic Regression, Decision Tree).
- Evaluate using accuracy, precision, recall, and F1-score.

5. Interpretation

- Identify the most important features.
- Visualize decision boundaries or use explainable outputs (e.g., SHAP, optional).

Deliverables

- A Jupyter notebook with all steps clearly commented
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